

2 (a) pressure = force / area (normal to the force) [clear ratio essential] B1 [1]

(b) (i) $P = mg / A = (5.09 \times 9.81) / A$ C1

$$A = (\pi d^2 / 4) = \pi \times (9.4 \times 10^{-2})^2 / 4 (= 0.00694 \text{ m}^2) \quad \text{C1}$$

$$\begin{aligned} P &= 49.93 / 0.00694 \\ &= 7200 (7195) \text{ Pa (minimum of 2 s.f. required)} \end{aligned} \quad \text{A1 [3]}$$

(ii) $\Delta P / P = \Delta m / m + 2\Delta d / d$ C1

$$= 0.01 / 5.09 + (2 \times 0.1) / 9.4 (= 0.0020 + 0.021 \text{ or } 2.3\%) \quad \text{C1}$$

$$\Delta P = 170 (165 \text{ to } 167) \text{ Pa} \quad \text{A1 [3]}$$

(iii) $P = 7200 \pm 200 \text{ Pa}$ A1 [1]